

AI Usage Disclosure

Inside Airbnb — New York City | Supplementary Deliverable (Assignment Section 10)

In accordance with Section 10 of the assignment brief, this document provides full transparency on AI tool usage throughout the assignment: which tools were used, where, how outputs were validated, and which AI suggestions were rejected or substantially modified. It mirrors Appendix A of the main report as a stand-alone artifact.

A.1 AI Tools Used

Tool	Model / Version	Usage
Claude (Anthropic)	Claude Sonnet 4.6	Primary AI assistant — code architecture advice, data-quality-framework design, business-interpretation drafting, SQL query review, and report-writing assistance.
ChatGPT (OpenAI)	GPT-4o	PRIMARY — Alternative perspective on statistical methodology, cross-checking pipeline design decisions, and generating initial boilerplate code templates.
Gemini (Google)	Gemini 1.5 Pro	PRIMARY — Data exploration queries, validation of EDA interpretations, and research on NYC Local Law 18 regulatory context.
GitHub Copilot	GPT-4-based	SUPPORTING — Inline code completion within the Jupyter environment. Used for boilerplate pandas and matplotlib code.
Cursor	Claude/GPT-4 backend	SUPPORTING — AI-assisted code editing for refactoring the ingestion pipeline and restructuring functions for modularity.
Perplexity	Online LLM	SUPPORTING — Quick factual lookups: Inside Airbnb methodology, NYC regulatory environment, and DuckDB SQL syntax references.

A.2 AI-Assisted Sections

Section	Assistance Level	Details
3.1 — Ingestion Pipeline	Medium	Claude suggested the retry / exponential back-off pattern and the MD5-checksum approach to incremental processing. The logic was implemented and tested independently.
3.1.3 — Quality Scoring	High	Claude suggested the structure of the quality_report() function. The completeness formula and grading thresholds were set independently, based on domain judgment.
3.1.4 — Fuzzy Duplicate Detection	Low–Medium	ChatGPT suggested using SequenceMatcher from Python's difflib library. The grouping logic and confidence tiers were designed independently.
3.4 — Star Schema Design	Medium	Claude confirmed the proposed star-schema structure was appropriate. DuckDB-specific implementation was done independently.

3.5 — Metadata Management	Medium–High	Claude assisted with the SQLite schema design for pipeline metadata. The architectural choice to separate operational from analytical concerns was the candidate's own.
4 (EDA) — Business Interpretations	Medium–High	Claude provided draft interpretations for the geographic pricing gradient, host segmentation, and review-score inflation findings. Each was reviewed, fact-checked, and substantially revised.
11 — Business Recommendations	Medium	Claude helped structure the recommendations table format. All substantive recommendations were derived from the analysis itself, not generated by AI.
This Report	Medium	Claude assisted with section structure, prose drafting, and table formatting. All quantitative claims were verified against notebook output before inclusion.

A.3 Key Prompts Used (Illustrative)

Prompt Summary	Tool	Purpose
<i>I have a pandas DataFrame with 90 columns and 30K rows representing Airbnb listings. Design a quality scoring function that measures completeness, validity, uniqueness, and consistency. Return a single score out of 100.</i>	Claude	Generated the initial quality_report() function structure.
<i>Given this correlation matrix [...], which features are most important price predictors and what are the confounds I should be aware of?</i>	ChatGPT	Assisted with statistical interpretation of the correlation matrix.
<i>I found that 30+ night minimums dominate the NYC calendar. What regulatory context explains this?</i>	Claude + Perplexity	Confirmed NYC Local Law 18 as the explanatory mechanism for the 30-night minimum-stay finding.
<i>I need to implement incremental processing so the pipeline only re-downloads files when the source has changed. What is the most reliable approach?</i>	Claude	Claude recommended MD5 checksums; the _is_changed() function and metadata ledger concept came from this response.
<i>I want to segment Airbnb hosts into casual vs. professional operators. What is the best way to define the cut-off point?</i>	Gemini	Gemini suggested the literature-standard 2+ listings threshold; independently validated against the NYC data distribution.
<i>Write a plain-English interpretation of a geographic pricing gradient finding: listings closer to Times Square have higher prices (r = -0.31 with distance).</i>	Claude	Claude's draft was substantially revised — the original 'strong' label was corrected to 'moderate' given r = -0.31.
<i>What is a good star schema structure for an Airbnb listing analysis database?</i>	Claude + ChatGPT	Both tools suggested similar structures; candidate chose DuckDB over suggested PostgreSQL. dim_date was added independently.

A.4 Output Validation

Every AI-generated output — whether code, statistical interpretation, or prose — went through one or more of the following validation steps before being adopted into the submission:

Code Validation

- All AI-generated code was executed in the actual Jupyter notebook environment against the real NYC Inside Airbnb dataset.
- Output values were compared against manually computed spot-checks.
- Unusual or unexpected outputs were investigated rather than accepted.

Statistical and Quantitative Validation

- Every number stated in the main report was verified against the actual notebook output. For example, the correlation coefficient $r = -0.31$ was verified against the `scipy.stats.pearsonr()` output before being cited.
- AI-stated correlation interpretations were cross-checked against standard social-science interpretation guides ($r < 0.3$ = weak, $0.3-0.5$ = moderate, > 0.5 = strong). Claude's original 'strong' label for $r = -0.31$ was corrected to 'moderate'.
- Review score statistics (median 4.75–4.94) were verified against the `listings_enriched.describe()` output before business interpretations were written.

Business Interpretation Validation

- AI-drafted business interpretations were checked against the computed data — if an interpretation implied a trend the data didn't support, it was revised or removed.
- The NYC Local Law 18 regulatory context was verified against publicly available NYC government documentation, not just the AI response.
- Host segmentation percentages (74% casual, 26% professional) were independently computed from the listings DataFrame before citing them.

Report-Level Validation

- Before finalizing the DOCX report, each section was read against the corresponding notebook output to confirm internal consistency.
- All table values in the Executive Summary were traced back to specific notebook cells.
- ChatGPT was used to review the Executive Summary for clarity — the feedback was treated as one input, not the final version.

A.5 Cases Where AI Suggestions Were Rejected or Modified

AI Suggestion	Decision	Reason
<i>Use PostgreSQL for the data warehouse</i>	Rejected — used DuckDB instead	PostgreSQL requires server setup; DuckDB is zero-server and integrates directly with notebooks. Better fit for the assignment context.
<i>Use ARIMA for seasonal price forecasting</i>	Deferred — not implemented	Calendar data lacks a price column in this scrape, making time-series price forecasting impossible. Implementing ARIMA against missing data would be methodologically unsound.
<i>Described the $r = -0.31$ correlation as a 'strong' finding</i>	Modified — described as 'moderate'	$r = -0.31$ is a weak-to-moderate correlation. Describing it as 'strong' would overstate the finding. The corrected interpretation notes that location matters but is not the dominant price driver.
<i>Use Unicode bullet characters directly in notebook markdown</i>	Modified — used proper list syntax instead	Unicode bullets cause rendering inconsistencies across Jupyter environments.
<i>Segment by 5+ listings as the commercial operator threshold</i>	Modified — 2+ listings threshold used	Academic literature (Wachsmuth & Weisler 2018) uses 2+ listings as the standard cut-off. Independently validated against the NYC data distribution.
<i>Use Streamlit for an interactive dashboard</i>	Deferred — time-constrained	Given the assignment's guidance to prioritize depth over breadth, the time investment in Streamlit would have reduced quality of core data engineering and EDA sections.

<i>ChatGPT suggested adding 'beds' as a top price predictor</i>	Rejected — kept but flagged for multicollinearity	beds, bedrooms, and accommodates are highly correlated (VIF > 10). The candidate retained accommodates as the representative feature.
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A.6 Summary Statement

Principles Followed

- **Intentionality:** AI tools were used with a specific purpose for each interaction, not as a general search engine.
- **Verification:** Every AI output that influenced the submission was independently validated before inclusion.
- **Transparency:** AI assistance is documented at the section level, not just mentioned generically.
- **Critical Judgment:** AI suggestions that conflicted with domain knowledge, statistical principles, or engineering best practices were rejected and the reasoning documented.